Appendix to: The Reality Gap in Automated Driving Systems Testing: Landscape, Challenges, and Research Directions

STEFANO CARLO LAMBERTENGHI, Technical University of Munich; fortiss GmbH, Germany
OLIVER WEIßL, Technical University of Munich; NorCom GmbH & Co. KGaA, Germany
JOHANNES BETZ, Technical University of Munich, Germany
ANDREA STOCCO, Technical University of Munich; fortiss GmbH, Germany

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A APPENDIX

A.1 Primary-Study Mappings

The tables in this appendix give the complete mapping from the categories of the corresponding Results tables to their primary studies.

A.1.1 RQ_1: Context.

A.1.2 $RQ_{2.2}$: Mitigation Techniques.

Table 1. Complete mapping of $RQ_{2.2}$ mitigation families and techniques to primary studies. A study can appear under several families and techniques.

Mitigation technique	Studies	Primary studies
Real-equipment integration (ViL/HiL/MR/XiL)	56	
Vehicle-in-the-loop (on-road)	28	[8, 11, 26, 29, 31, 37, 40, 41, 82, 89, 114, 115, 117, 130, 135, 138, 152, 158, 160, 161, 164, 165, 178, 181, 217, 232–234]
Mixed reality	21	[9–11, 24, 26, 29, 37, 84, 89, 114, 115, 138, 152, 158, 160, 161, 177, 199, 217, 232, 234]
Controller/ECU hardware-in-the-loop	6	[19, 25, 28, 32, 40, 46]
Co-simulation	5	[18, 114, 159, 164, 185]
Vehicle-in-the-loop (testbench)	4	[24, 71, 87, 102]

Authors' addresses: Stefano Carlo Lambertenghi, lambertenghi@fortiss.org, Technical University of Munich; fortiss GmbH, Boltzmannstraße 3, Garching, Bavaria, Germany, 85748; Oliver Weißl, Technical University of Munich; NorCom GmbH & Co. KGaA, Boltzmannstraße 3, Garching, Bavaria, Germany, 85748; Johannes Betz, Technical University of Munich, Boltzmannstraße 15, Garching, Bavaria, Germany, 85748; Andrea Stocco, Technical University of Munich; fortiss GmbH, Boltzmannstraße 3, Garching, Bavaria, Germany, 85748.

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Mitigation technique	Studies	Primary studies
Camera sensor-in-the-loop	3	[25, 144, 188]
Human-in-the-loop	3	[21, 24, 51]
Physical scale-model / surrogate-platform validation	2	[157, 197]
LiDAR sensor-in-the-loop	1	[139]
Modular X-in-the-loop framework	1	[40]
Radar sensor-in-the-loop	1	[231]
Small-scale vehicle-in-the-loop	1	[14]
Steering-actuator hardware-in-the-loop	1	[92]
XiL/HIL	1	[61]
Generative content synthesis	41	
Diffusion-based generation	14	[13, 22, 50, 62, 80, 113, 128, 151, 174, 200–202, 218, 229]
GAN-based generation/translation	12	[6, 59, 97, 133, 134, 137, 150, 166, 182, 184, 186, 220]
Neural rendering (NeRF/3DGS)	6	[30, 72, 110, 183, 214, 228]
Synthetic data generation/augmentation	4	[10, 103, 183, 215]
Synthetic perturbation/augmentation	3	[69, 203, 219]
Image compositing	1	[223]
Physics-based multi-sensor object insertion	1	[57]
Point-cloud/image reprojection & warping	1	[5]
Traffic/agent behaviour modelling	29	
Learning-based (imitation/RL) traffic models	7	[4, 5, 17, 23, 211, 216, 225]
Traffic/behaviour-model calibration	5	[99, 111, 125, 204, 209]
Route/trajectory-conditioned behaviour modelling	4	[109, 124, 176, 222]
Learned/generative trajectory models	3	[56, 206, 221]
Rule-based/behaviour-tree traffic models	3	[100, 116, 141]
Optimization/search-based generation	2	[78, 86]
Statistical/distributional traffic calibration	2	[60, 213]
Field-test/traffic-flow calibration	1	[3]
Human-reactive traffic models	1	[21]
Traffic/NPC behaviour modelling	1	[33]
Real-world evidence reconstruction	28	
Recorded driving/sensor-log reconstruction	12	[18, 35, 91, 93, 112, 123, 142, 176, 180, 190, 194, 226]
Crash/accident-evidence reconstruction	6	[15, 54, 205, 210, 224, 230]
Map/aerial-derived reconstruction	5	[1, 108, 146, 155, 175]
Scenario-generation framework/toolchain	2	[18, 77]
Data-driven simulator construction	1	[162]
Domain adaptation	1	[12]
Reconstruction-strategy comparison (classical vs. neural assets)	1	[120]
Retrieval-augmented domain adaptation	1	[235]
Sensor physics modelling	28	
LiDAR sensor modelling	14	[7, 27, 35, 52, 65, 73, 79, 88, 120, 121, 139, 147, 154, 189]
Camera sensor modelling	6	[35, 45, 76, 104, 157, 191]

Mitigation technique	Studies	Primary studies
Radar sensor modelling	5	[66, 118, 119, 192, 231]
Camera+LiDAR+radar sensor modelling	2	[70, 95]
Compute/timing-fidelity emulation	2	[75, 203]
Ego policy/control transfer	13	
Reinforcement-learning-based policy transfer	5	[34, 64, 74, 112, 219]
Vehicle-dynamics/controller calibration	5	[2, 38, 44, 55, 85]
Imitation-learning-based policy transfer	2	[4, 5]
Generative-video-assisted policy transfer	1	[202]

A.1.3 $RQ_{3.3}$: Evaluation Metrics.

Table 2. Complete mapping of $RQ_{3.3}$ evaluation-metric categories and specific metrics to primary studies. A study can appear under several metrics.

Metric category / metric	Studies	Primary studies
Sensor/perception comparison	107	
Detection accuracy/rate	30	[10, 12, 14, 15, 27, 35, 43, 66, 76, 104, 120, 121, 131, 132, 139, 145, 147, 149, 154, 156, 157, 163, 168, 183, 189, 191, 198, 199, 214, 215]
Average precision (mAP/AP)	27	[6, 16, 27, 50, 59, 62, 66, 76, 81, 103, 104, 120, 121, 128, 139, 140, 171, 179, 183, 191, 212, 214, 215, 220, 224, 229, 235]
IoU/segmentation overlap	20	[9, 47, 52, 65, 96, 103, 121, 126, 137, 158, 171, 179, 183, 184, 199, 214, 215, 224, 228, 235]
Precision/recall/F1	19	[14, 16, 19, 27, 37, 59, 74, 76, 81, 103, 105, 120, 128, 144, 145, 150, 171, 184, 215]
Distributional image/video realism (FID/FVD/KID)	18	[22, 30, 65, 128, 133, 134, 166, 171, 174, 183, 199, 200, 202, 214, 218, 220, 228, 229]
Structural similarity (SSIM)	15	[10, 30, 70, 76, 103, 110, 128, 144, 158, 171, 183, 199, 214, 227, 228]
RMSE/MAE-family error	13	[6, 10, 52, 65, 103, 105, 139, 144, 168, 171, 199, 214, 227]
Learned perceptual similarity (LPIPS)	12	[30, 103, 110, 128, 133, 158, 171, 199, 214, 218, 228, 229]
Signal fidelity (PSNR)	11	[10, 30, 72, 76, 110, 158, 171, 183, 199, 214, 228]
Correlation coefficient	9	[6, 8, 34, 46, 52, 70, 118, 168, 189]
Distribution-distance test	8	[47, 65, 79, 118, 126, 127, 168, 171]
Point-cloud distance (Chamfer/point-to-point/Minkowski)	8	[16, 65, 79, 126, 127, 196, 199, 215]
Domain-adaptation/perception-transfer performance	6	[25, 45, 69, 97, 106, 151]
LiDAR return-rate/intensity accuracy	6	[7, 72, 88, 149, 199, 214]
Point-cloud/feature distributional distance (EMD/Wasserstein)	6	[47, 79, 126, 127, 168, 171]
Sensor field-of-view/detection-range (hardware coverage, not driving behaviour)	6	[52, 67, 83, 139, 154, 189]
Learned/perceptual feature comparison	5	[22, 110, 183, 188, 200]
Sensor/signal-measurement difference	4	[66, 67, 119, 158]
Classification accuracy	3	[149, 179, 204]
Material/reflectance-property fidelity	3	[27, 95, 154]
Pixel-level error	3	[65, 144, 171]
Radar cross-section (RCS) accuracy	3	[83, 118, 192]
Sensor-model fidelity validation (acceptance/qualification criteria)	3	[73, 149, 187]
Signal degradation/artifact score (blur/noise/contrast)	3	[148, 149, 188]
Regression fit/error statistic	2	[67, 105]
Texture/hand-crafted feature metric	2	[42, 43]
3D-scene reconstruction depth/consistency error (RMSE/MAE/iMAR/iRMSE)	1	[226]
DNN scenario-matching error (image similarity)	1	[173]
Feature-embedding distance	1	[53]

Metric category / metric	Studies	Primary studies
GAN-generated image-realism distributional comparison	1	[182]
Human/object 3D pose-estimation accuracy	1	[132]
Image/perception-level distributional quality metrics	1	[97]
ML-testing coverage metrics (neuron coverage/surprise adequacy/ fault-revealing ability)	1	[6]
Occupancy-grid comparison	1	[147]
Procedurally-reconstructed environment coverage/fidelity	1	[155]
Sensor/signal-measurement difference (radar)	1	[231]
Set-overlap distance (Jaccard/Hamming)	1	[189]
Variance/standard deviation	1	[6]
ADS behaviour/trajectory comparison	96	
Speed/velocity comparison	36	[2, 3, 14, 20, 21, 31, 32, 36, 39–41, 48, 55, 58, 62–64, 67, 72, 83, 86, 99, 112, 113, 130, 138, 164, 168, 176, 181, 190, 193, 201, 213, 221, 230]
Trajectory error (ADE/FDE/RMSE/ATE)	15	[17, 40, 55, 56, 85, 87, 120, 134, 138, 142, 176, 202, 209, 214, 225]
Trajectory/path point-wise distance (sim vs. real)	13	[4, 17, 20, 23, 49, 58, 63, 94, 96, 108, 130, 141, 180]
RMSE/MAE-family error	11	[2, 36, 40, 48, 55, 86, 168, 199, 209, 214, 216]
Lane-keeping/car-following behaviour	9	[74, 99, 107, 108, 112, 156, 164, 168, 203]
Distribution-distance test	6	[23, 98, 99, 113, 168, 176]
Inter-vehicle/relative-spacing distance	5	[48, 62, 138, 176, 201]
Real-vs-simulated driving-behaviour comparison	5	[5, 142, 151, 161, 216]
Trajectory-shape distance (DTW/Hausdorff/Fréchet/LCSS)	5	[85, 100, 113, 199, 221]
Path-tracking control performance	4	[2, 92, 138, 197]
Variance/standard deviation	4	[4, 32, 39, 113]
Correlation coefficient	3	[48, 168, 209]
Multi-object tracking accuracy	3	[57, 98, 145]
Trajectory similarity/distance score	3	[75, 96, 123]
Interaction/traffic-behaviour modelling accuracy	2	[109, 225]
Planning/route-control outcome	2	[120, 175]
Trajectory variance/disengagement/validity rate	2	[4, 86]
Waymo Sim Agents realism benchmark score	2	[62, 80]
ADAS-function pass/fail (LKA/ACC/AEB)	1	[156]
Composite multivariate scenario-scoring metric (collection of sub-metrics)	1	[167]
Copula-based driver-interaction model fit (reconstruction/simulation accuracy, MSE)	1	[60]
Crash-scenario reconstruction accuracy/coverage	1	[54]
DNN driving-decision scenario-matching error (steering-angle similarity)	1	[68]
Driver-behaviour distributional comparison	1	[217]
Endpoint/waypoint offset error	1	[158]
Full-stack co-simulation toolchain end-to-end validation outcome	1	[18]
Generated-scenario kinematic-validity check (acceleration/jerk within physical limits)	1	[113]
Generated-trajectory diversity & realism score (FID/MMD)	1	[22]
Generated-trajectory diversity & realism score (fidelity/diversity/ reactivity/feasibility)	1	[222]
Inter-vehicle gap and traffic-flow distribution comparison	1	[111]
LLM-as-judge scenario-realism robustness score	1	[208]
LLM-as-judge scenario-realism robustness/success rate	1	[207]
Localization/pose-tracking accuracy	1	[87]
Moving-object trajectory reproduction accuracy (vs. real ground truth)	1	[204]
Multi-vehicle tracking + traffic-flow reconstruction accuracy	1	[198]
Per-model simulation-reliability quantitative score	1	[101]

Metric category / metric	Studies	Primary studies
Point-wise trajectory comparison (raw positions)	1	[142]
Policy-training success rate/loss	1	[74]
Policy/traffic-behaviour descriptive measures	1	[219]
QD scenario-generation coverage/sample-efficiency	1	[78]
Regression fit/error statistic	1	[67]
Scenario-description quality & coverage	1	[93]
Trajectory comparison over simulation time	1	[159]
Trajectory-reconstruction optimization fit/convergence	1	[123]
Trajectory/kinematic-state value list (raw, per timestep)	1	[130]
Trajectory/path adaptation performance (simulation vs. field test)	1	[84]
VIL-platform signal-curve reliability metrics (Geers/Zilliacus)	1	[129]
Vehicle dynamics/actuation comparison	37	
Acceleration	17	[20, 21, 23, 36, 44, 48, 62, 99, 112, 113, 138, 164, 168, 176, 190, 206, 221]
Steering	8	[32, 55, 64, 92, 130, 138, 164, 172]
Jerk	6	[23, 57, 113, 206, 221, 225]
RMSE/MAE-family error	6	[36, 40, 44, 48, 55, 168]
Distribution-distance test	5	[23, 99, 113, 168, 176]
Braking/deceleration	4	[3, 143, 158, 181]
Yaw	4	[21, 57, 130, 164]
Correlation coefficient	3	[44, 48, 168]
Energy/fuel-consumption & traffic-flow comparison	3	[3, 63, 112]
Variance/standard deviation	3	[32, 39, 113]
Vehicle-state tracking error	3	[21, 40, 172]
Bare 'vehicle dynamics' comparison (combinations of ACC/AEB/LKA)	1	[156]
Longitudinal-control tracking performance (speed/following distance)	1	[31]
Signal-level RMSE/NRMSE (sim vs. real vehicle-dynamics signals)	1	[136]
Significance/confidence testing	1	[206]
Vehicle-dynamics model-mismatch metric (ν-gap/terrain-gap/ internal-dynamics-gap)	1	[195]
Vehicle-dynamics parameter-estimation RMSE	1	[38]
Velocity/spacing mean & standard deviation	1	[39]
Safety/event outcome	34	
Collision/crash rate	26	[11, 15, 17, 29, 33, 49, 61, 62, 74, 86, 113, 143, 156, 162, 168, 175, 176, 201, 203, 205, 206, 211, 221, 225, 228, 230]
Time-to-collision/PET	12	[33, 61, 62, 86, 113, 143, 162, 168, 201, 206, 221, 228]
Pass/fail or disengagement rate	8	[8, 33, 162, 169, 170, 172, 221, 228]
Safety distance/near-miss	3	[44, 165, 176]
Scenario criticality score (composite risk indicators)	1	[77]
Traffic-safety distributional statistics	1	[213]
No metric reported	23	[24, 28, 51, 71, 82, 89, 91, 102, 114–117, 124, 135, 152, 153, 177, 178, 185, 186, 210, 223, 232]
Runtime/computational	19	
Memory/compute utilization	7	[13, 52, 75, 110, 128, 163, 218]
Latency/runtime	6	[27, 32, 57, 74, 125, 158]
Frame rate (FPS)	3	[52, 110, 140]
Inference time	2	[19, 128]
Network/communication load	2	[160, 233]
Accuracy-vs-compute tradeoff (Pareto efficiency)	1	[163]
Model training time	1	[235]
ODD-coverage & ADS-failure testing overhead	1	[13]
Round-trip time (RTT)	1	[39]
Qualitative assessment	16	

Metric category / metric	Studies	Primary studies
Visual/qualitative inspection	11	[26, 35, 70, 81, 94, 146, 172, 192, 194, 219, 234]
Likert-scale rating	3	[1, 220, 225]
Human/user study	2	[90, 140]
Whiteness/subjective realism score	1	[140]
Human/participant behavioural comparison	1	[122]

A.1.4 $RQ_{4.1}$: Simulation Platforms and Reference Datasets.

Table 3. Complete mapping of $RQ_{4.1}$ simulation platforms and reference datasets to primary studies. A study can name several platforms and datasets.

Simulation platform / dataset	Studies	Primary studies
Simulation platforms	182	
CARLA	45	[8, 13, 19, 24, 25, 32, 34, 40, 41, 43, 51, 53, 62, 64, 65, 82, 89, 92, 98, 100, 103, 104, 114, 116, 126, 127, 131, 133, 134, 137, 146, 150, 159, 163, 166, 180, 186, 188, 200, 203, 210, 224, 225, 227, 235]
IPG CarMaker	16	[18, 20, 35, 67, 104, 106, 119, 124, 129, 130, 142, 145, 164, 169, 185, 193]
Unity	16	[37, 39, 52, 63, 71, 79, 81, 86, 96, 139, 160, 177–179, 221, 229]
MATLAB/Simulink	13	[20, 21, 44, 58, 67, 119, 136, 139, 142, 164, 166, 185, 210]
SUMO	12	[3, 64, 99, 114, 123, 159, 164, 177, 178, 181, 216, 233]
Custom/research simulator	11	[10, 84, 111, 118, 135, 143, 162, 165, 183, 204, 205]
VIRES VTD	9	[27, 36, 76, 102, 138, 145, 167, 170, 191]
SVL/LGSVL	7	[49, 94, 113, 179, 206, 221, 230]
Gazebo	7	[28, 29, 112, 197, 219, 232, 234]
LidarSim	6	[73, 120, 121, 127, 214, 215]
Autoware simulator	6	[32, 55, 89, 139, 152, 179]
PreScan	4	[58, 68, 117, 185]
RoadRunner	4	[21, 82, 146, 190]
Donkeycar	4	[97, 140, 172, 173]
AutoDRIVE	3	[151–153]
CarSim	3	[62, 166, 231]
esmini	3	[48, 154, 180]
Siemens Simcenter	2	[2, 185]
nuPlan	2	[33, 78]
FMU co-simulation	2	[2, 61]
BeamNG	2	[6, 54]
Apollo simulator	2	[1, 85]
Blender/rendering engine	2	[30, 189]
VISTA	2	[4, 5]
Scenic scenario language	2	[90, 103]
Dataset/reference data	66	
KITTI	18	[6, 43, 57, 59, 79, 97, 110, 128, 137, 139, 150, 155, 171, 179, 188, 226, 228, 235]
nuScenes	10	[30, 35, 50, 57, 72, 103, 211, 218, 225, 228]
Cityscapes	9	[47, 59, 110, 137, 166, 191, 220, 224, 229]
Waymo Open Dataset	9	[30, 62, 72, 80, 103, 110, 179, 202, 228]
Virtual KITTI	8	[6, 43, 53, 97, 128, 155, 171, 228]
Custom/study-collected dataset	7	[23, 93, 97, 125, 173, 213, 229]
Traffic-trajectory dataset (NGSIM/ HighD/rounD/INTERACTION)	7	[105, 123, 180, 201, 209, 216, 222]
PandaSet	4	[72, 199, 214, 228]
GTA-V synthetic dataset	3	[43, 47, 59]

Simulation platform / dataset	Studies	Primary studies
Udacity driving dataset	2	[6, 68]
Other named dataset	2	[33, 215]
BDD100K	2	[98, 223]
Randomized/procedurally-generated driving dataset	1	[69]
XLD dataset	1	[107]
UTDT benchmark dataset	1	[198]
MS-COCO	1	[9]
CARLA-collected dataset	1	[133]
AutoDRIVE-collected dataset	1	[151]
Duckietown Gym dataset	1	[184]
DeepScenario dataset	1	[207]
Custom LiDAR-simulation dataset	1	[7]
Crash/accident-event dataset	1	[131]
Argoverse	1	[176]
No simulation platform or reference dataset reported	19	[11, 15, 16, 22, 26, 42, 60, 83, 88, 91, 115, 144, 148, 168, 174, 194, 196, 212, 217]

A.1.5 $RQ_{4.2}$: Hardware Platforms.

A.1.6 $RQ_{4.3}$: Systems Under Test.

Table 4. Complete mapping of $RQ_{4.3}$ systems under test and specific technologies to primary studies. A study can appear under several systems.

System under test	Studies	Primary studies
No SUT	89	[1, 13, 14, 21–23, 29, 36, 38, 42, 43, 51, 53, 56, 60, 62, 63, 65, 67, 70, 72, 77, 80, 88, 90, 91, 93–95, 99–101, 103, 105, 107–110, 117, 122, 123, 127, 146, 148, 151, 155, 159, 160, 162, 166, 167, 170, 174–178, 180, 182, 183, 185–187, 189, 191, 192, 194, 196, 198, 200, 201, 204, 205, 209–213, 216–218, 222–224, 226, 227, 230, 232, 233]
Camera perception	40	
YOLO-family detector	15	[6, 35, 37, 47, 59, 97, 98, 128, 131, 144, 150, 163, 171, 220, 229]
Object detector	12	[10, 30, 35, 45, 69, 76, 104, 145, 163, 179, 199, 214]
Segmentation model	7	[30, 47, 121, 133, 137, 184, 199]
BEV-based detector/feature extractor	4	[16, 50, 134, 235]
Lane detection	3	[97, 134, 172]
VRU (pedestrian/cyclist) detection & driver-warning system	3	[24, 50, 157]
Faster R-CNN detector	2	[9, 10]
3D human pose estimation	1	[132]
ADAS smart front camera (sensor and embedded software)	1	[25]
Depth estimation	1	[188]
Object detection + tracking	1	[214]
Full ADS	37	
Modular	29	[8, 11, 18, 19, 28, 32, 48, 49, 55, 71, 85, 89, 96, 102, 113, 114, 129, 130, 139–141, 152, 153, 156, 164, 181, 203, 206, 221]
End-to-end	8	[5, 33, 74, 96, 116, 140, 225, 228]
Black-box AV (architecture undisclosed)	1	[86]

System under test	Studies	Primary studies
Simplified driving function with PID vehicle controller (path following, TTC-based braking)	1	[41]
Ego-vehicle control/actuation	36	
Lane keeping/changing control	11	[4, 68, 97, 124, 125, 135, 164, 168, 172, 173, 219]
Adaptive cruise control	10	[3, 26, 31, 44, 61, 112, 124, 164, 168, 234]
Automatic emergency braking	6	[15, 46, 58, 82, 135, 143]
Steering control	4	[54, 92, 134, 142]
Vehicle control stack/ECU	3	[32, 143, 169]
Model predictive control	2	[2, 20]
Ground-robot dynamics control system	1	[195]
Integrated trajectory-planner + vehicle controller	1	[40]
PID trajectory-tracking controller	1	[39]
Rule-based reactive control policy	1	[78]
Waypoint-following control	1	[136]
Planning/trajectory prediction	18	
Path/motion planning	14	[28, 34, 40, 48, 55, 75, 84, 120, 158, 161, 164, 197, 202, 214]
Decision-making/planning policy	3	[28, 64, 111]
Trajectory/motion prediction	2	[17, 214]
Automated valet-parking planner	1	[138]
LiDAR perception	17	
3D object detection	7	[12, 27, 35, 120, 121, 149, 215]
PV-RCNN	3	[12, 79, 139]
BEV feature extractor (LiDAR/camera fusion)	2	[16, 215]
LiDAR object detection + tracking	2	[147, 154]
PointPillars	2	[27, 81]
Sensor simulation/generative model	2	[73, 106]
LiDAR intensity simulation model	1	[7]
Point-cloud segmentation	1	[121]
PointNet(++)	1	[52]
PolarNet	1	[52]
RangeNet++	1	[52]
VoxelNet	1	[79]
Radar perception	7	[66, 83, 118, 119, 126, 193, 231]
Localization	5	
Modular-ADS localization component	2	[28, 129]
GNSS/GPS-based localization	1	[190]
NDT-based mapping/localization	1	[139]
SLAM-based localization	1	[87]
LLM/foundation-model reasoning agent	3	[165, 207, 208]
Multi-sensor fusion perception/ tracking	2	[57, 115]

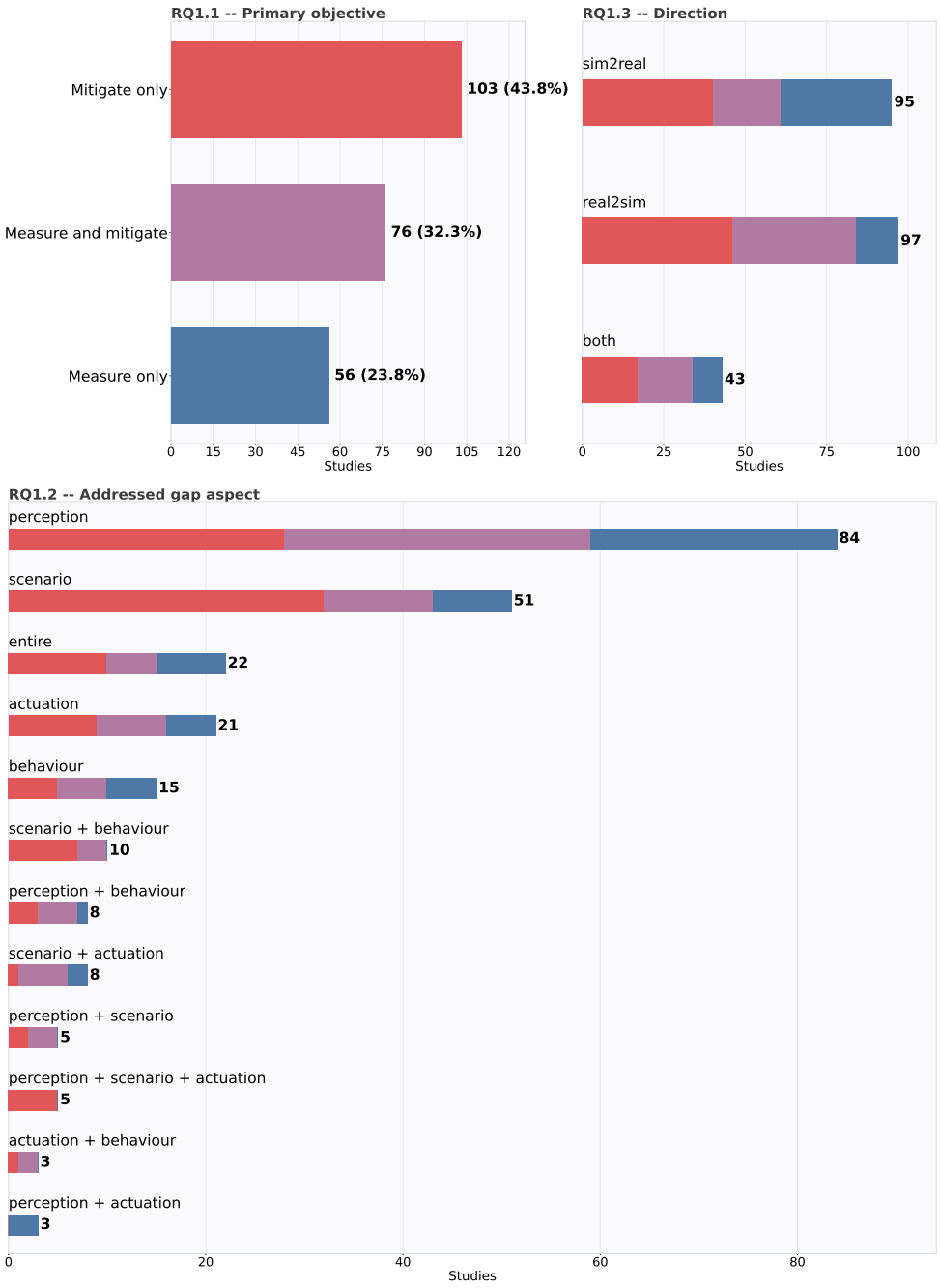


Fig. 1. $RQ_{1.1}$ primary objective (left), $RQ_{1.3}$ direction (right), and $RQ_{1.2}$ addressed gap aspect (bottom).

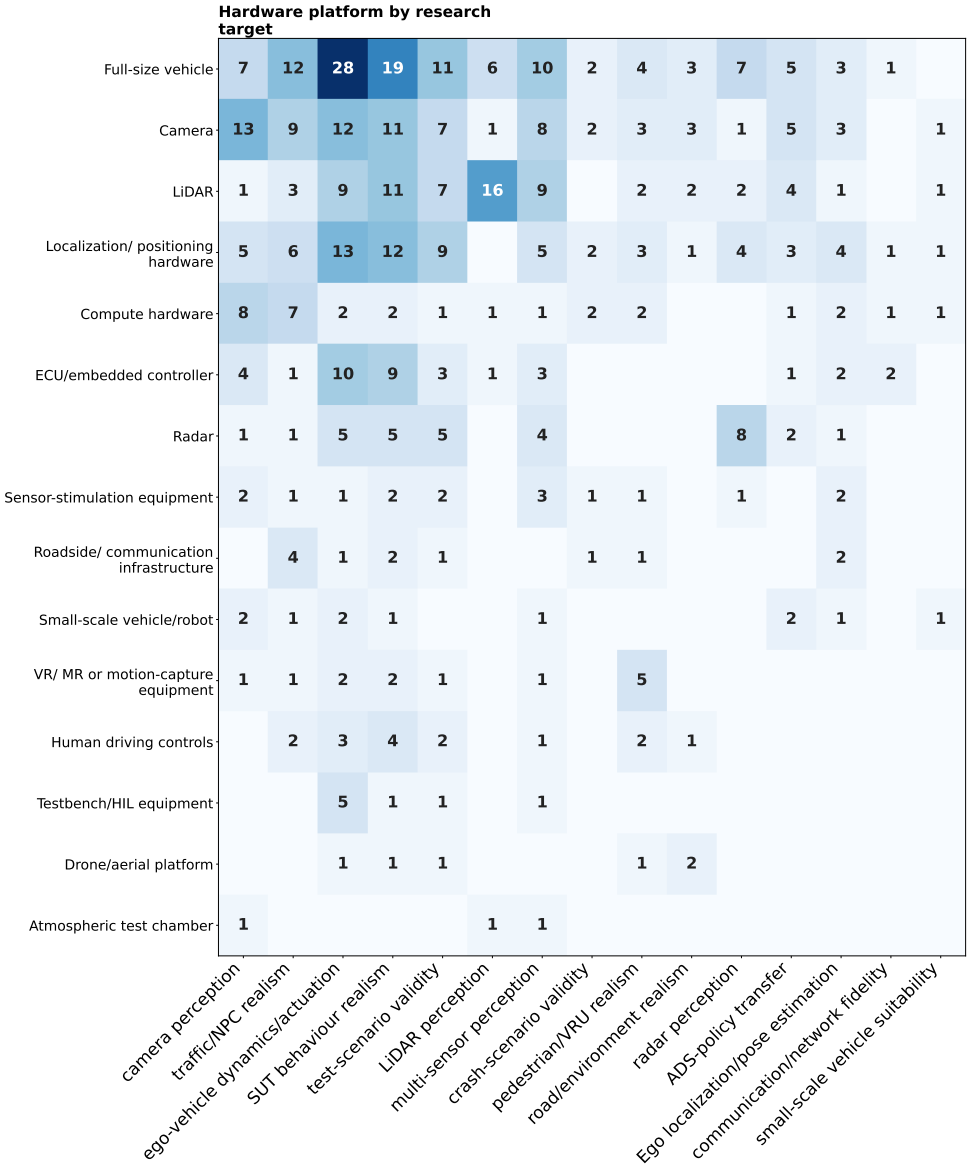


Fig. 2. $RQ_{4.2}$ hardware platforms by research target.

A.2 Mitigation Studies by Execution Profile

Table 5 lists the mitigation studies in each of six mutually exclusive execution profiles, from fully simulated to fully real.

Table 5. Mitigation studies by execution profile. Profiles are mutually exclusive.

Category	Subcategory	Citations
Execution profile	Fully simulated	[13, 52, 55, 64, 66, 71, 73, 75, 116, 118, 128, 147, 192, 200, 203]
	Calibrated simulation	[1, 3, 4, 6, 7, 10, 15, 17, 18, 22, 23, 30, 33, 35, 45, 50, 54, 56, 57, 59, 60, 62, 65, 72, 76–78, 80, 86, 93, 97, 99, 100, 103, 110–113, 117, 121, 123–125, 133, 137, 141, 150, 154, 155, 159, 166, 174–176, 180, 183–186, 190, 191, 199, 201, 202, 204–206, 209–211, 213–216, 218, 220–225, 228–230, 235]
	Real-data reuse	[12, 91]
	Single-component physical integration	[19, 27–29, 32, 38, 40, 44, 61, 69, 70, 74, 79, 85, 119, 120, 134, 142, 146, 181, 182, 188, 189, 219, 231, 234]
	Multi-component physical integration	[2, 5, 8, 9, 11, 14, 21, 24–26, 31, 34, 37, 41, 46, 51, 82, 84, 87, 89, 92, 102, 104, 108, 109, 114, 130, 135, 138, 139, 144, 151, 152, 157, 158, 160–162, 164, 165, 177, 178, 194, 197, 217, 226, 232, 233]
	Fully real	[88, 95]
	Outside all six profiles (described individually in the text)	[115]

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